KARTHIKEYAN THIAGARAJAN

Senior Lead Software Engineer | Frontend Architect (Angular / React / TypeScript)

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PROFESSIONAL SUMMARY

Senior Lead Software Engineer with **15+ years of experience** architecting and delivering **enterprise-scale web applications** across banking, SaaS, IoT, and digital platforms. Deep expertise in **Angular 16/17, React, and TypeScript**, with a proven record of leading frontend architecture, building reusable component libraries, and shipping high-performance, compliant UIs for global enterprises including **Citibank**. Hands-on with **CI/CD, Docker, and DevOps tooling**, and experienced in integrating **LLMs (Claude, GPT, Llama 3, Qwen3, Mistral)** into production applications using **LangChain, LlamaIndex, Ollama, and RAG pipelines**.

CORE SKILLS

Frontend: Angular 16/17, React JS, TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, Angular Material, RxJS, Responsive Design, Component Architecture, State Management, Performance Optimization, Accessibility (WCAG)

Backend & Integration: REST API Integration, Node.js, Express.js, JSON, Async Data Handling, GraphQL (basic)

AI / LLM Engineering: LLM Integration (Claude, GPT, Llama 3, Qwen3, Mistral), LangChain, LlamaIndex, Ollama, RAG Pipelines, Prompt Engineering

DevOps & Quality: CI/CD Pipelines, Docker, Git, GitHub, Bitbucket, SonarQube, Code Reviews, Agile/Scrum

Architecture & UX: Frontend Architecture, Reusable Component Libraries, Design Systems, Enterprise UI Standards, UX Optimization

PROFESSIONAL EXPERIENCE

Career Break — **May 2026 – Present**

- Currently available for new opportunities.

Senior Lead Software Engineer | Virtusa Consulting Services (Client: Citibank) — **May 2023 – May 2026**

Project: Citi Resource & Financial Management Platform (Banking Domain)

Tech: *Angular 17, TypeScript, SCSS, Angular Material, REST APIs, .NET services, Lightspeed CI/CD, Docker*

- Led **frontend architecture** for Citi's enterprise resource and financial management platform used for **headcount planning, budget forecasting, approvals, and governance workflows**.
- Designed and built scalable Angular components powering **10+ business modules** — Position Management, Headcount Dashboards, Payments, Accruals, and Software Renewals.
- Implemented **multi-stage approval workflows** with audit trails, role-based access control (RBAC), and banking compliance controls.
- Optimized UI performance and responsiveness for enterprise-scale user loads; enforced code quality via SonarQube and peer reviews.
- Delivered through **Lightspeed CI/CD pipelines** with Docker-based builds across Dev, UAT, and Prod environments.
- Partnered with backend, DevOps, and business stakeholders in an Agile environment to drive releases end to end.

Senior Digital Innovation Engineer | Buckman Laboratories (India) Pvt Ltd — **May 2021 – Mar 2023**

Tech: *Angular, React, Node.js, Express.js, .NET, SQL Server, TypeScript, HTML5, CSS3*

- Built modern, scalable UI features in **Angular and React** for enterprise digital innovation platforms.
- Developed backend services and REST APIs using **Node.js, Express.js, and .NET** to support full-stack feature delivery.
- Improved UX consistency by designing a library of **reusable UI components** adopted across multiple applications.
- Delivered responsive, performance-optimized interfaces in fast-paced innovation teams, balancing speed with maintainable architecture.

Senior Frontend Developer | PushPro — **Jun 2020 – Mar 2021**

Tech: *JavaScript, HTML5, CSS3, Modern UI Frameworks*

- Developed **real-time, data-driven notification dashboards** and reusable UI components.
- Drove UI performance optimization, responsiveness, and usability improvements across the product.

Senior Frontend Developer | Auriss Technologies **Oct 2017 – Feb 2020**

Project: Akita Security & Compliance SaaS Platform
Tech: *JavaScript, HTML5, CSS3, UI/UX*

- Built enterprise-grade UI for a **security and compliance SaaS platform** serving enterprise customers.
- Contributed to scalable SaaS frontend architecture, component reuse, and responsive design standards.

UI / Web Developer | Pacific Controls, UAE **Feb 2015 – Oct 2016**

Project: IoT Smart Building Dashboards
Tech: *JavaScript, HTML5, CSS3, Responsive Design*

- Delivered **IoT-enabled dashboards** for smart building management with real-time data visualization and monitoring.
- Built high-performance, responsive interfaces for complex, data-heavy monitoring systems.

Front-End Web Developer | Webtek Solutions **Feb 2014 – Sep 2014**

- Developed Windows and web application UIs using XAML, LESS/CSS, and JavaScript; converted HTML templates into Angular views with responsive layouts.

Front-End Web Developer | Elcom **Oct 2012 – Jan 2014**

- Built enterprise UI components for ElcomCMS, a .NET-based content management platform powering corporate websites, portals, and content workflows.

Senior Associate, MarCom (e-Marketing Developer) | VWR International **Apr 2010 – Oct 2012**

- Developed promotional pages, e-newsletters, surveys, and localized digital content for European marketing teams across **20+ markets**.

Web Developer / UI Designer / Team Lead | Earlier Career (multiple companies) **2000 – 2009**

- Delivered web design, UI development, branding, and multimedia solutions at Almega Cable, Shriasys, Exceed International, Dot Com Infoway, and S3 Group.

PROJECTS

WingzAI Platform | Live wingzai.deltaphi.in

- Unified workspace to manage connectors, build **AI agents**, and automate security, compliance, and service workflows.

SigmaSec – Posture Intelligence Security | Prototype sigmasec.vercel.app

- Unified scanning with Nuclei, Trivy, and Gitleaks, plus exploitability intelligence and **AI-authored remediation**.

CogniScale | MVP cogniscale-eta.vercel.app

- Dynamic assessment and remediation platform built to replace rigid, one-size-fits-all testing, using **LLM and RAG**.

RAG System for Your Documents | Proof of Concept ask-mydocs.vercel.app

- Ask questions and retrieve insights from your own documents using a **Retrieval-Augmented Generation** system.

AI Agent Skills | Live carthworks.github.io/ai-agent-skills

- Drop-in SKILL.md files that give AI coding agents specialized, consistent behavior.

EDUCATION

B.Sc. Computer Science — Bharathiar University

KEY HIGHLIGHTS

- **15+ years** in frontend engineering and UI architecture, including **3 years leading frontend for a Tier-1 global bank (Citi)**.
- Expert in **Angular 16/17 and React** for large-scale, compliance-driven enterprise applications.
- Domain depth across **banking, SaaS, IoT, and security/compliance** platforms.
- Production experience integrating **LLMs and RAG pipelines** into web applications — a rare frontend + AI skill combination.
- CI/CD- and Docker-fluent engineer who works hand-in-hand with DevOps and backend teams.